

IT Research Methodology

Qualitative Data Analysis and Interpretation

Image Source: Research Methodology

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Outline

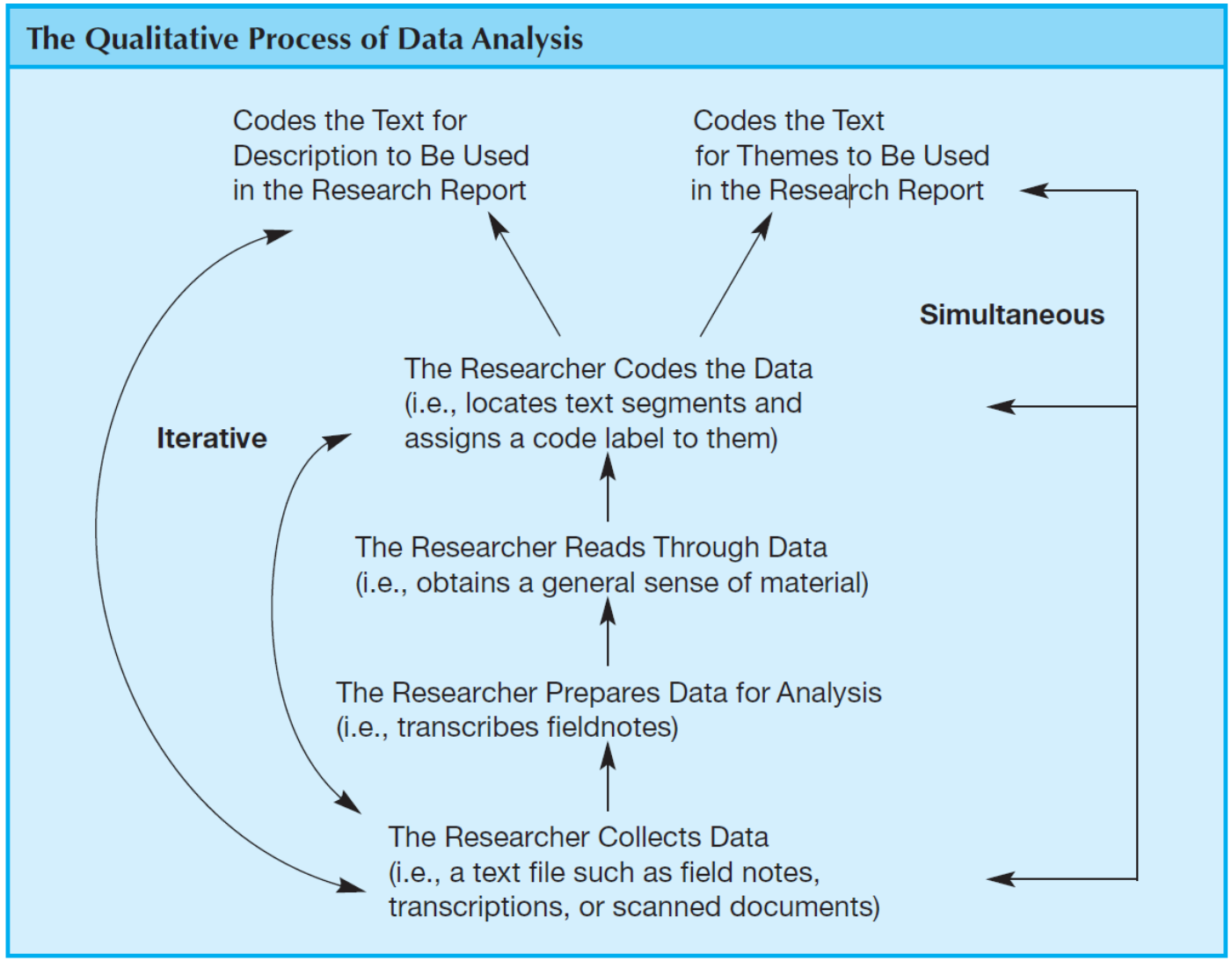
- Qualitative Process of Data Analysis
- Data Preparation and Organizing for Analysis
- Use of Qualitative Data Analysis Software Programs
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❖ Qualitative Process of Data Analysis

- Analyzing qualitative data requires understanding how to make sense of text and images so that you can form answers to your research questions
- Six steps commonly used in analyzing qualitative data:

1. Preparing and organizing the data for analysis
2. Engaging in an initial exploration of the data through the process of coding it
3. Using the codes to develop a more general picture of the data (descriptions and themes)
4. Representing the findings through narratives and visuals
5. Interpreting the meaning of the results by reflecting personally on the impact of the findings and on the literature that might inform the findings, and, finally
6. Conducting strategies to validate the accuracy of the findings

Qualitative Process of Data Analysis



❖ Data Preparation and Organizing for Analysis

➢ Initial preparation of the data for analysis:

- Organizing the vast amount of information,
- Transferring it from spoken or written words to a typed file
- Making decisions about whether to analyze the data by hand or by computer

❖ Data Preparation and Organizing for Analysis

➢ Organize Data:

- Developing a matrix or a table of sources that can be used to help organize the material
- Organizing the materials by type:
  - ✓all interviews,
  - ✓all observations,
  - ✓all documents, and
  - ✓all photographs or other visual materials;

  As an alternative, you might consider organizing the materials by participant, site, location, or some combination of these approaches

Data Preparation and Organizing for Analysis

- Keeping duplicate copies of all forms of data
- Maintaining participant confidentiality by storing data in password-protected electronic files or locked file drawers

❖ Data Preparation and Organizing for Analysis

➢ Transcribe Data:

- During qualitative data collection, you will collect text or words through interviewing participants or by writing field notes during observations
- Transcription is the process of converting audio recordings or field notes into text data.

❖ Data Preparation and Organizing for Analysis

➢ Analyze by Hand or Computer:

- Researchers have a choice about whether to hand-analyze data or to use a computer
- The **hand analysis of qualitative data** means that researchers read the data, mark it by hand, and divide it into parts.
- A hand analysis may be preferred in the following cases:
  - When you prefer to not use qualitative data analysis software
  - When you want to be close to the data and have a hands-on feel for it without the intrusion of a machine
  - When you have time to commit to a hand analysis, as it is a labor-intensive activity to manually sort, organize, and locate words in a text database

❖ Use of Qualitative Data Analysis Software Programs

- Qualitative data analysis software are programs that store data, organize it, enable you to assign labels or codes to your data, and facilitate searching through your data and locating specific text or words.
- Example: QualCoder

Use of Qualitative Data Analysis Software Programs

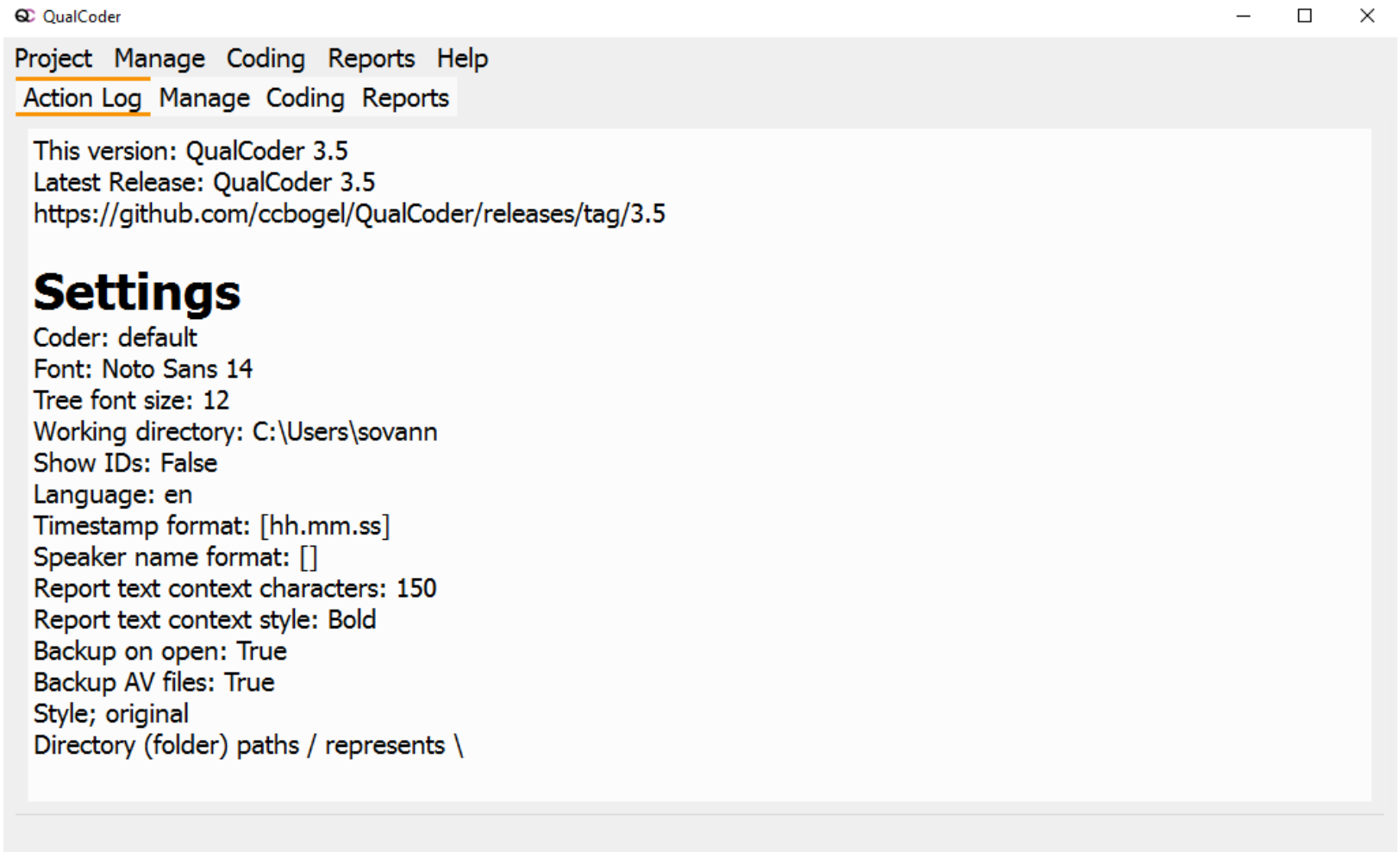
Hand-Coding a Page from a Sample Interview Transcript

The Process of Reconstructing Curriculum in a Rural High School Setting

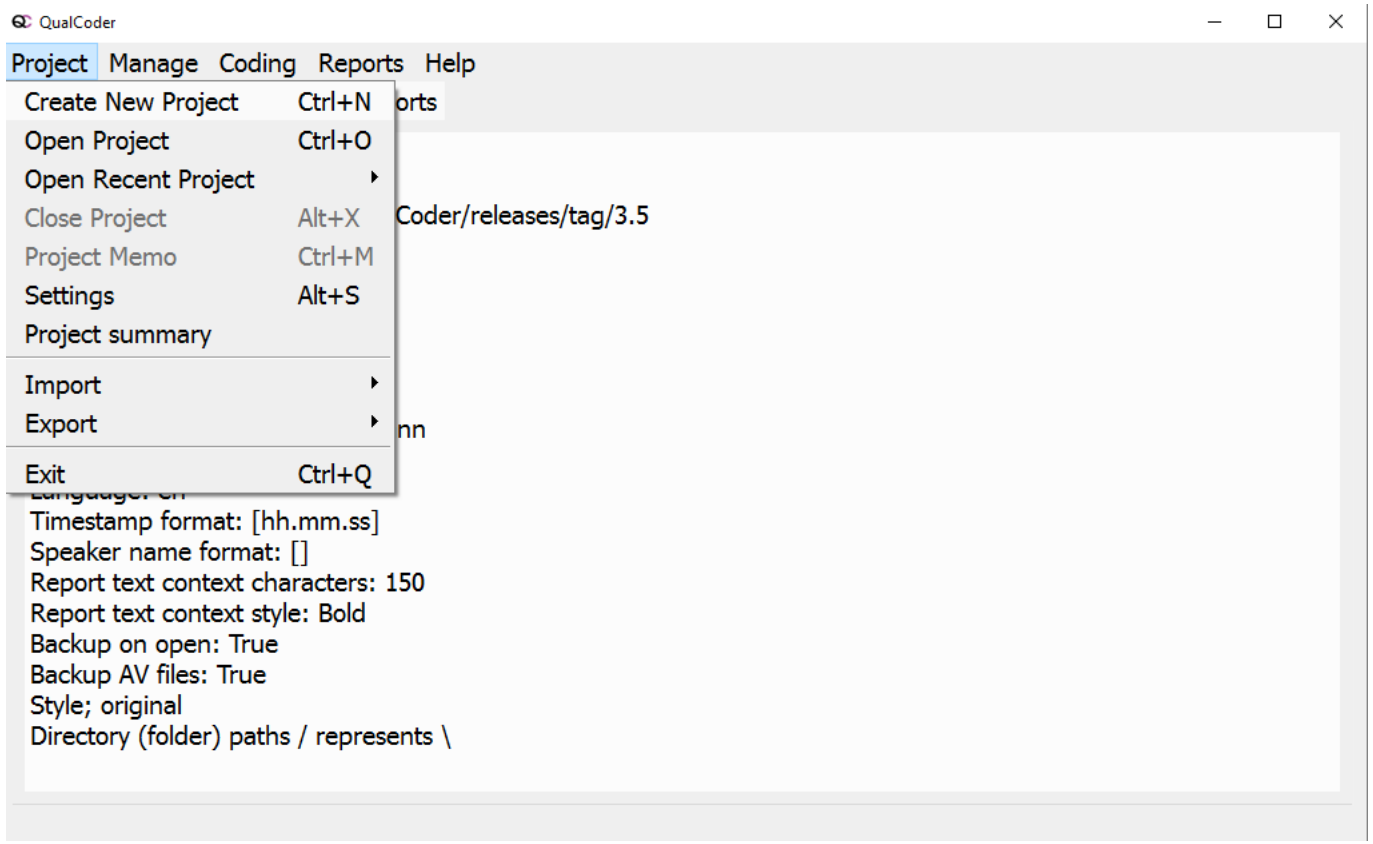
Codes Here		Themes (and Other Ideas) Here
	JJ: One thing, Lucy, that I've heard talked about was the fact that schools reflect the strengths of communities. What do you perceive as strengths of Greenfield as a community and how that relates to schools?	
Close-knit community	LU: Well, I think Greenfield is a fairly close-knit community. I think people are interested in what goes on. And because of that, they have a sense of ownership in the schools. We like to keep track of what our kids are doing and feel a connection to them because of that. The downside of that perhaps is that kids can feel that we are looking TOO close. But most of the time, that is the nurturing environment that we do provide an atmosphere of concern and	Potential theme: The community Idea: Getting a good sense here for the community and its values
Health of community or community values	care. To back up, you said the health of the community itself is reflected in schools. A lot of times communities look at schools and say they are not doing this or they aren't doing that, or we're missing something in our schools. I think perhaps we look at the school and see, this is probably a pretty conservative community overall, and look to make sure that what is being talked about in the schools really carries out the community's values. There is a little bit of an idealization I think, perhaps in terms of what we thought of "basic education." [And I think there might be a tendency to hold back a little bit too much because of that idealization of "you know, we learned the basics, the reading, the writing and the arithmetic."] So you know, any change is	A good quote
Change is threatening	threatening. And I think that goes for the community as well as what we see reflected at the school. Sometimes that can get in the way of trying to do different things. I think, again, idealization, older members of the community forget some of the immaturity that they experienced when they were in school and forgetting that kids are kids. So there is a little bit too much of that mental attitude. But for the most part, I think there is a sense of we're all in this together, and concern for the kids.	
	JJ: In terms of looking at leadership strengths in the community, where does Greenfield set in a continuum there with planning process, understanding the need to plan, forward-thinking, visionary people. You talked about that a little bit before.	Potential theme: Leaders
Visionary skills of talented people	LU: I think there are people that have wonderful visionary skills. I would say that the community as a whole would be . . . would not reflect that. I think there are people who are driving the process, but the rest of the community may be lagging behind a little bit. I think we have some incredibly talented people who become frustrated when they try to implement what they see as their . . .	Idea: Returns to description of community again

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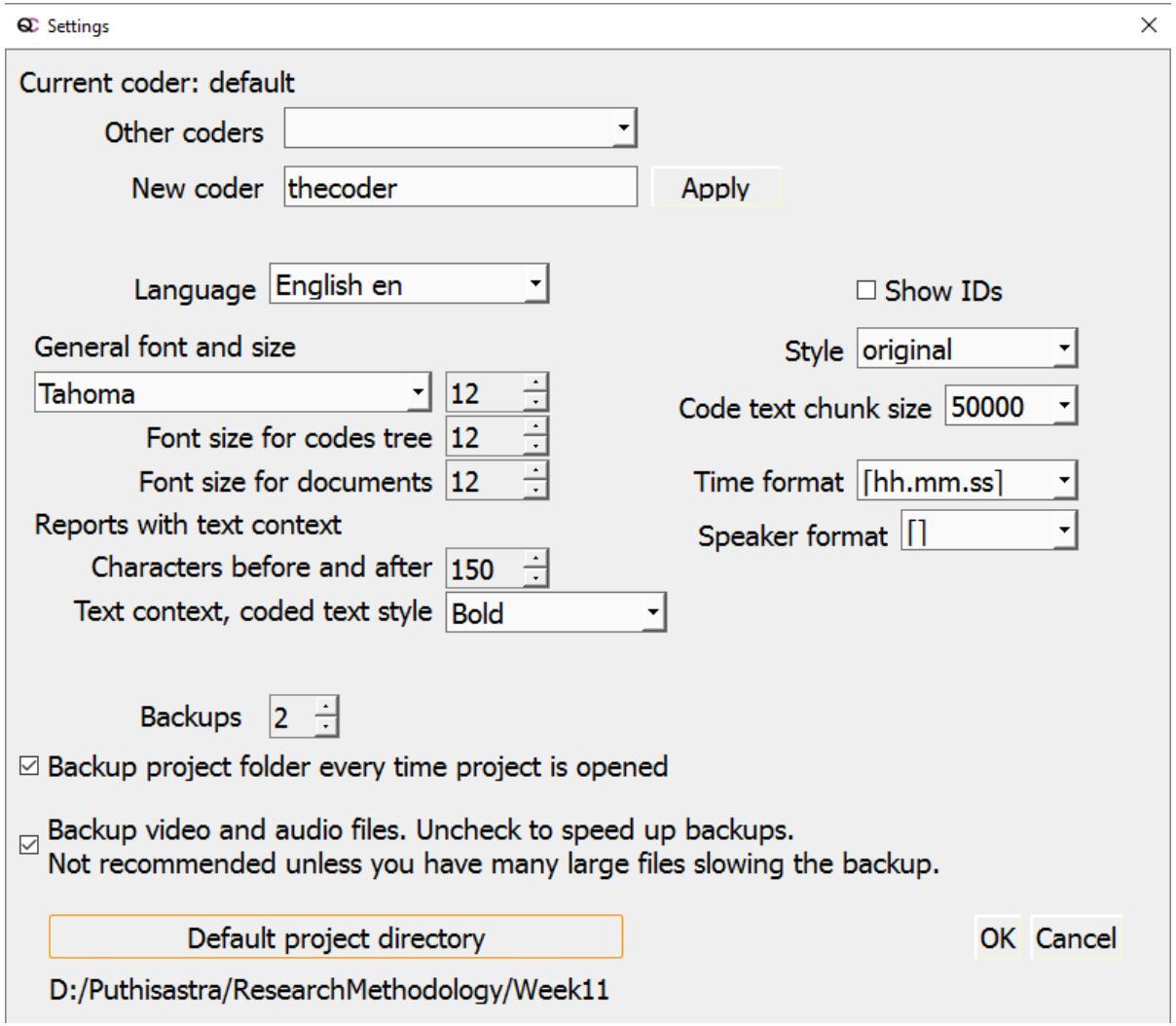
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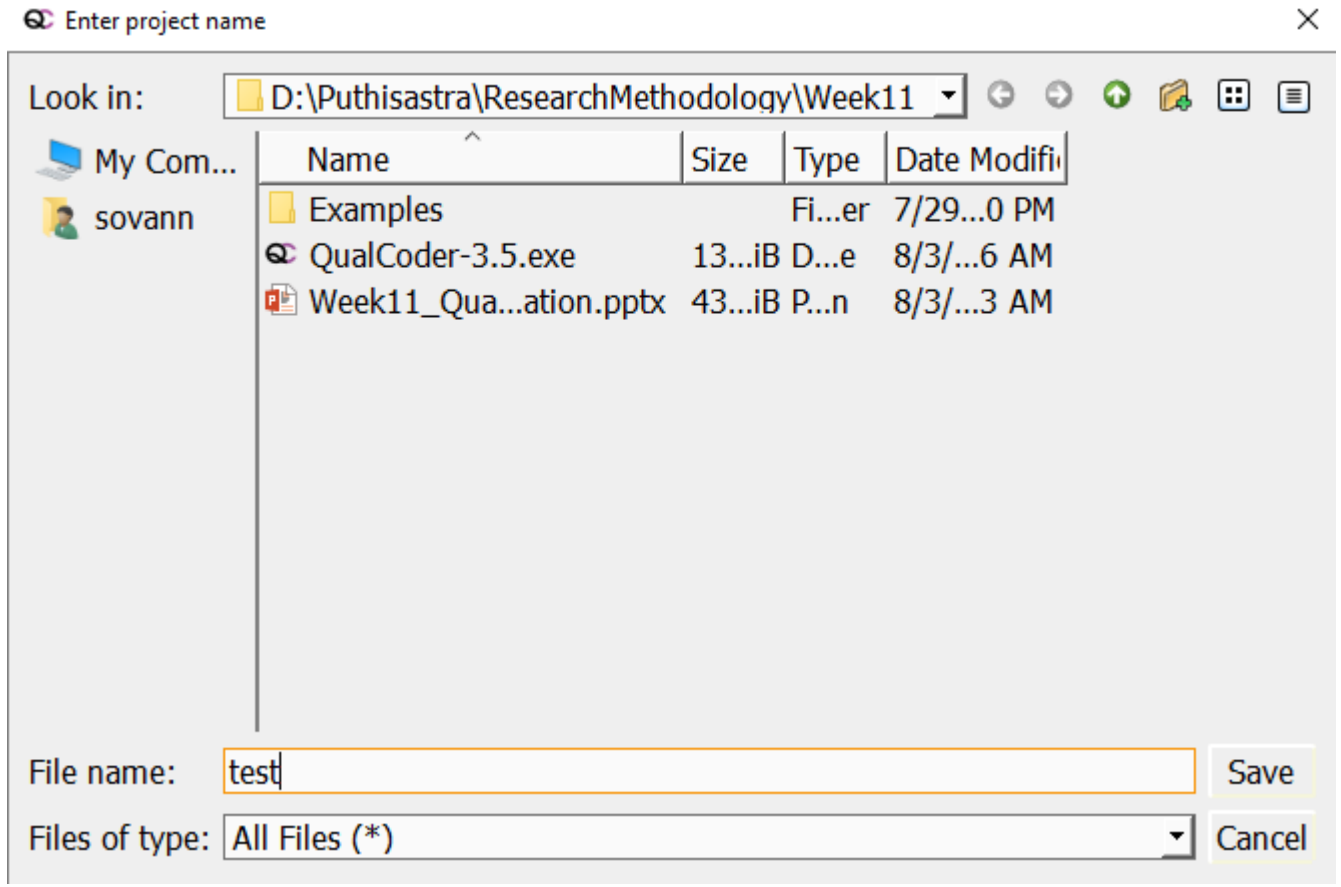
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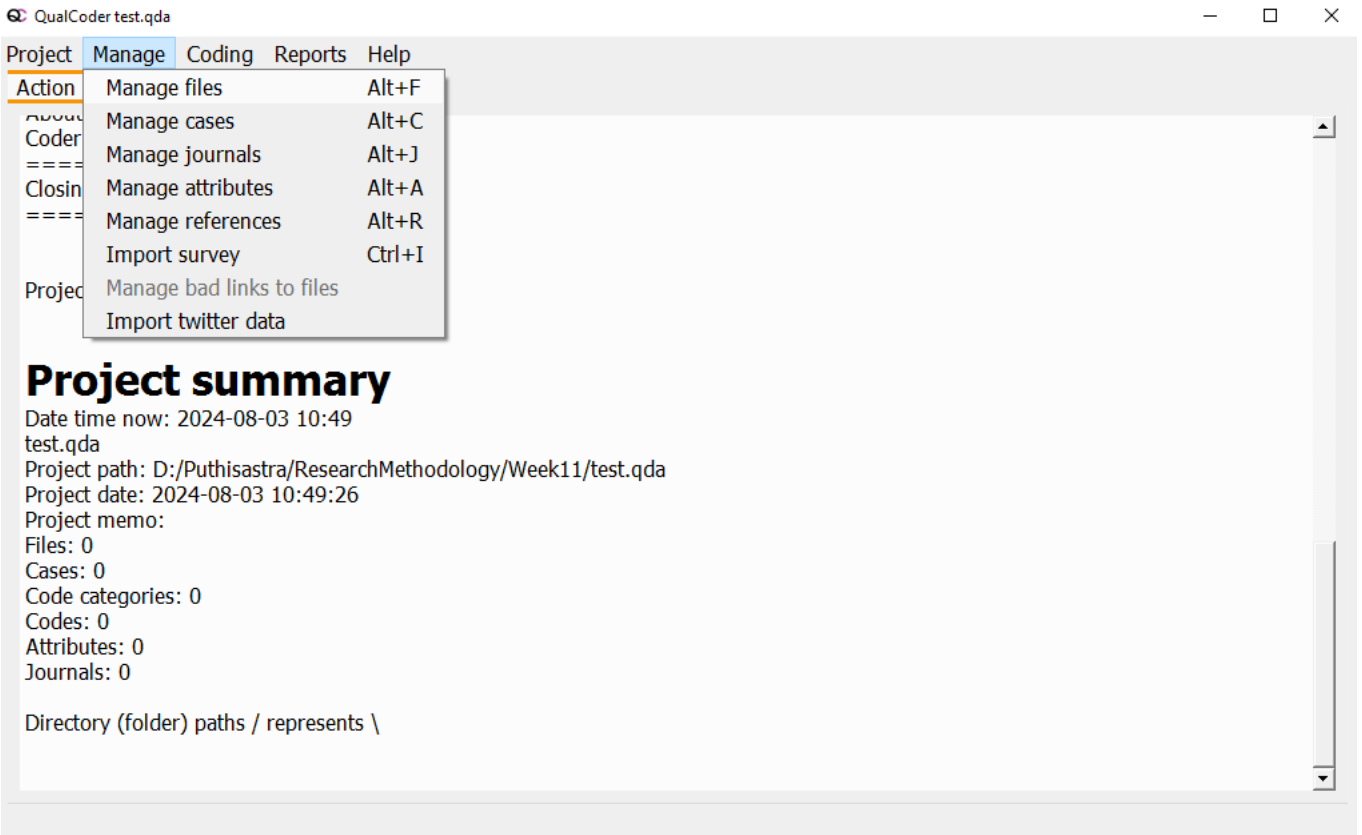
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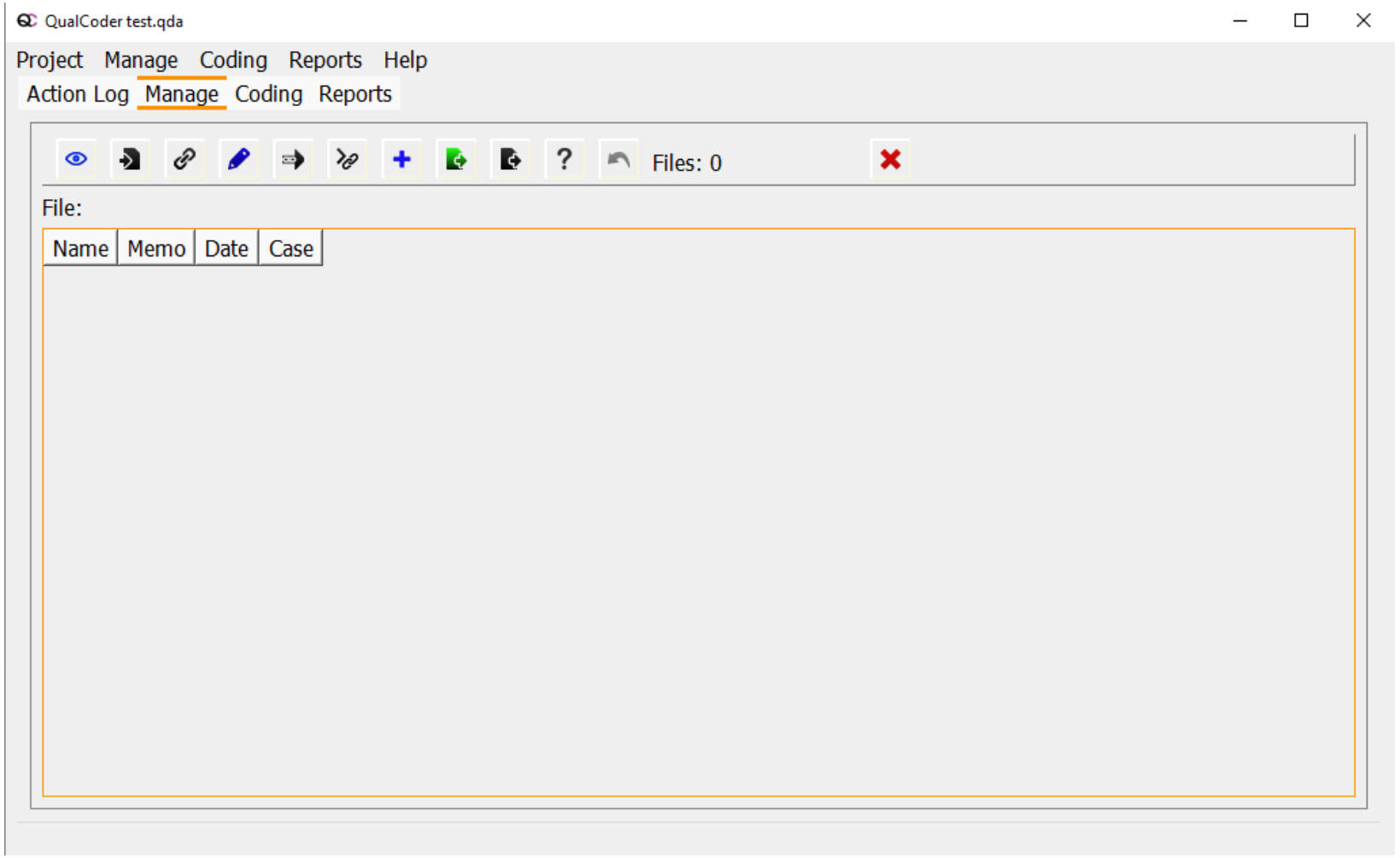
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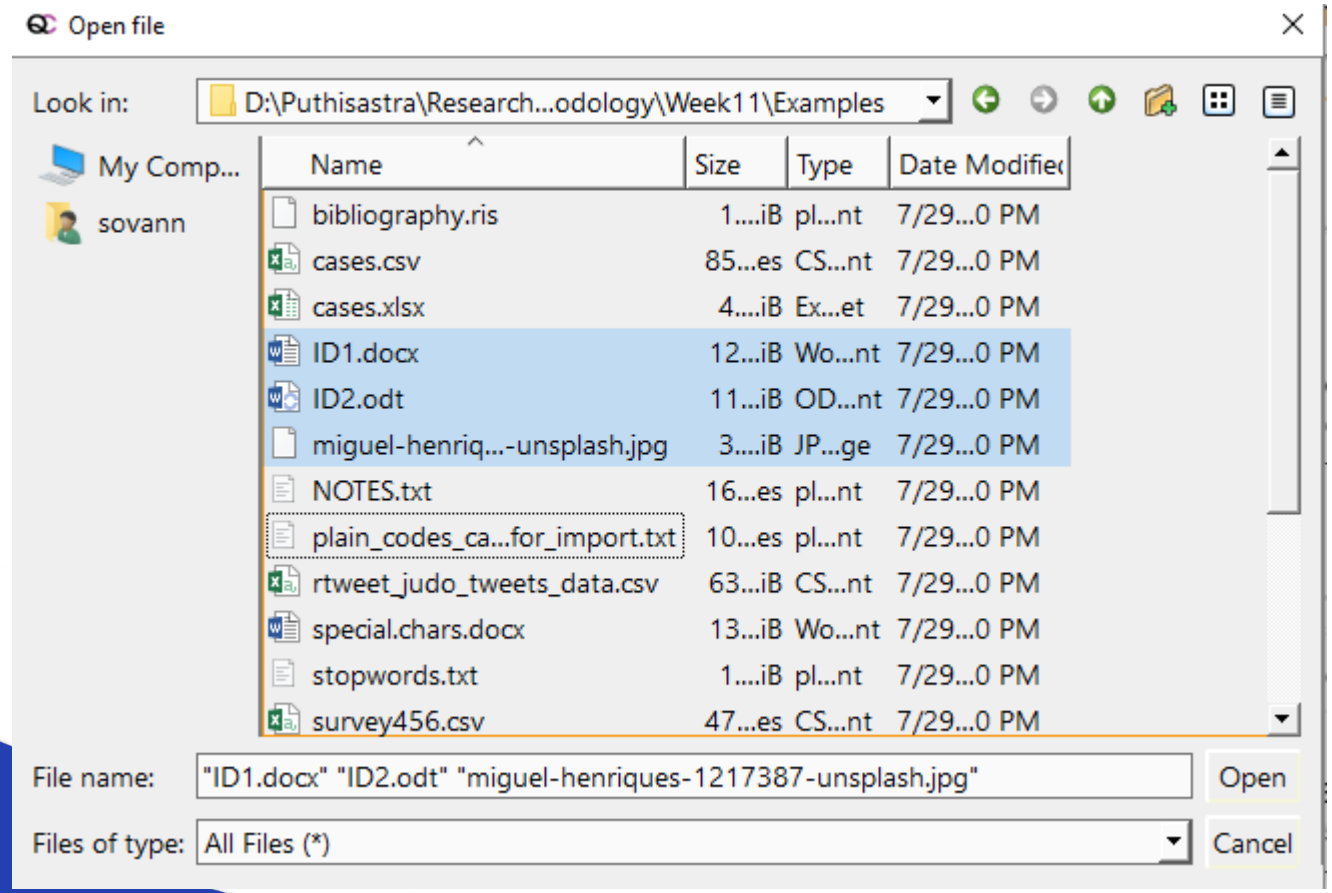
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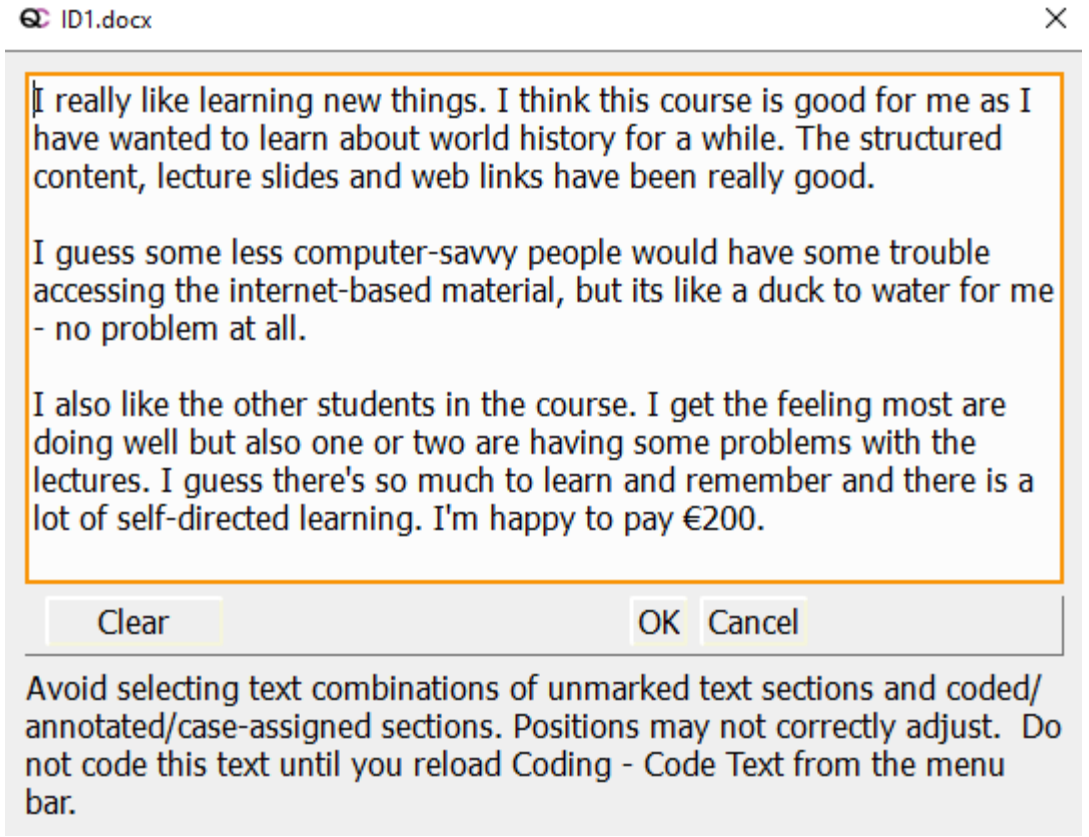


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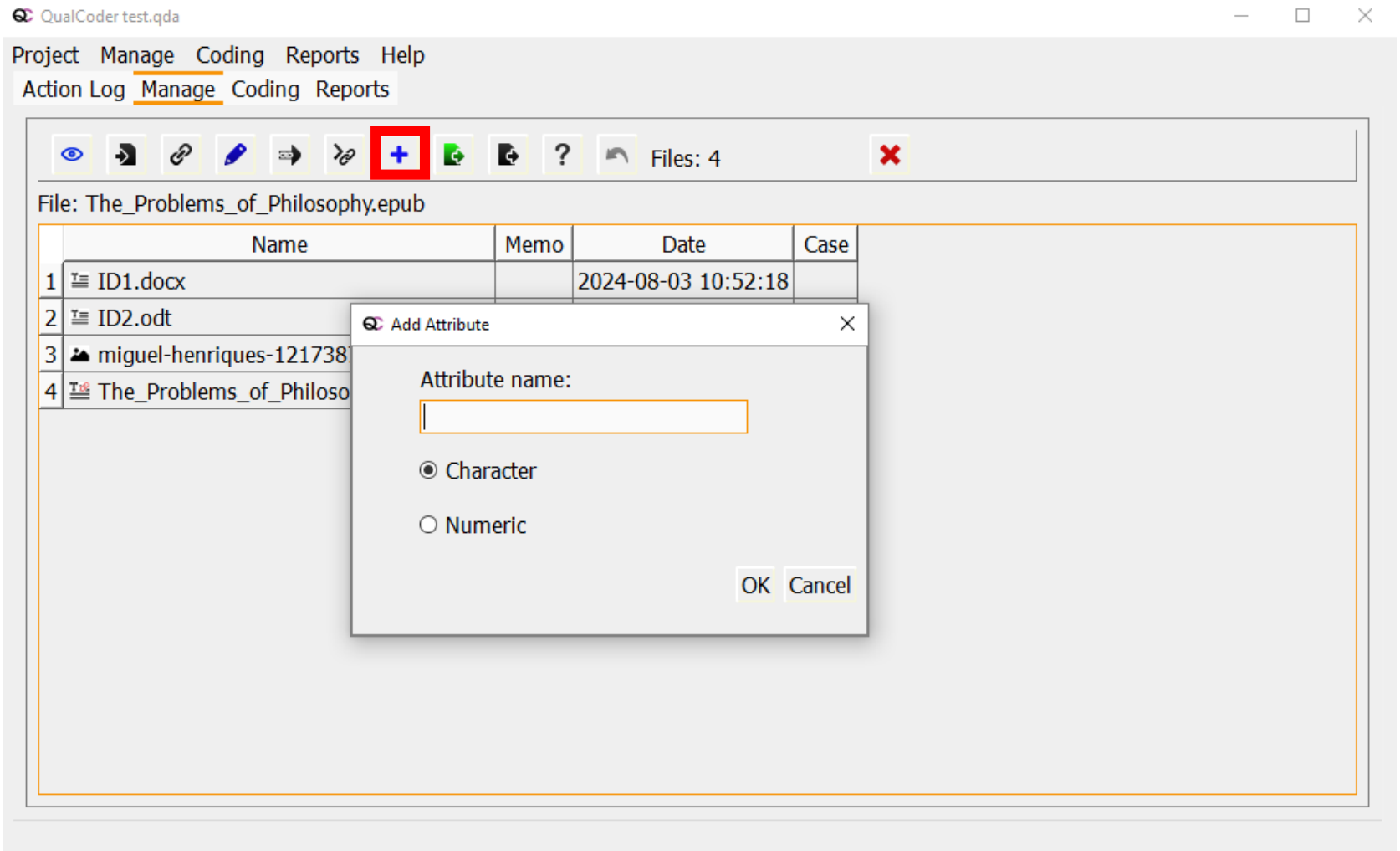


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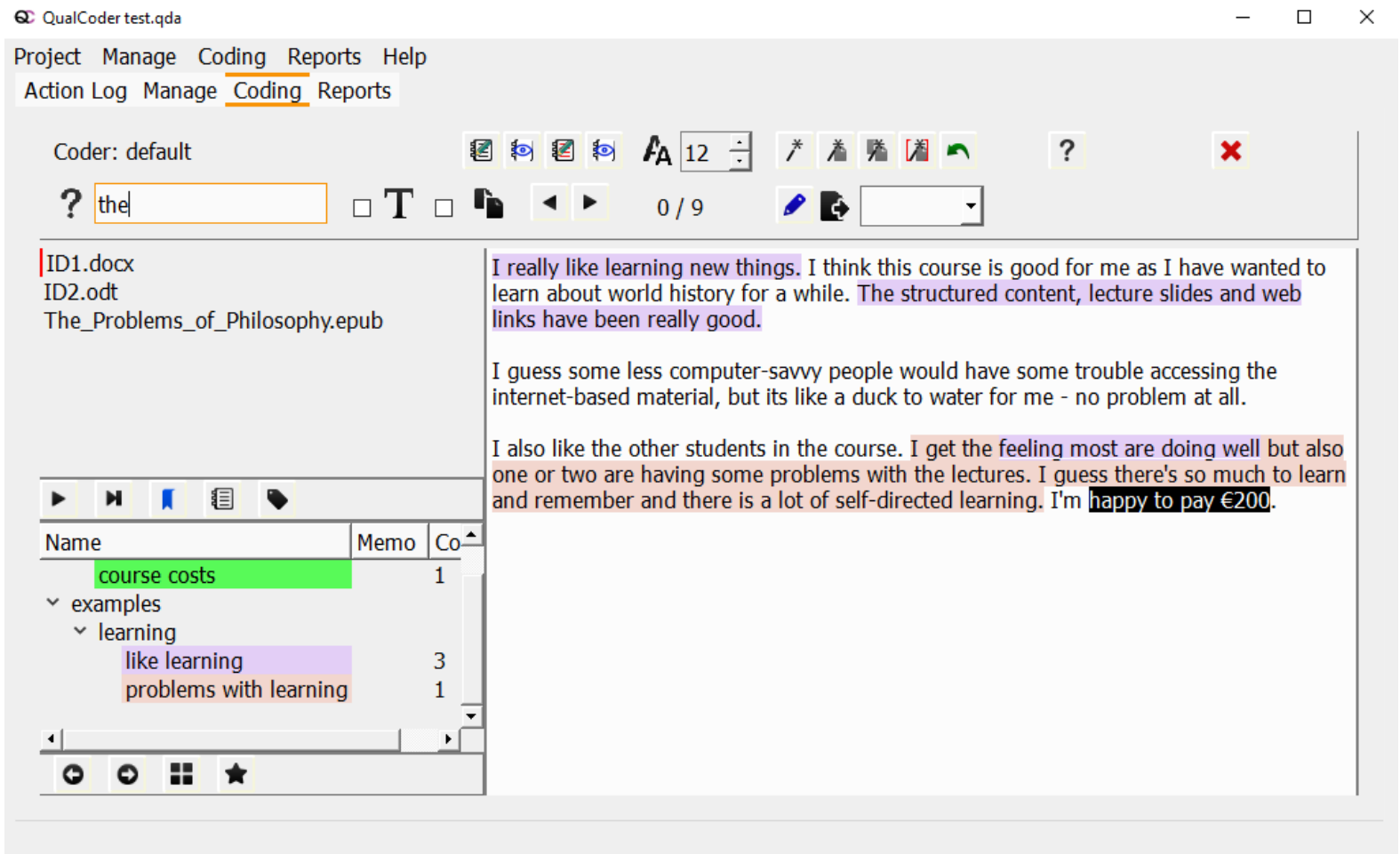




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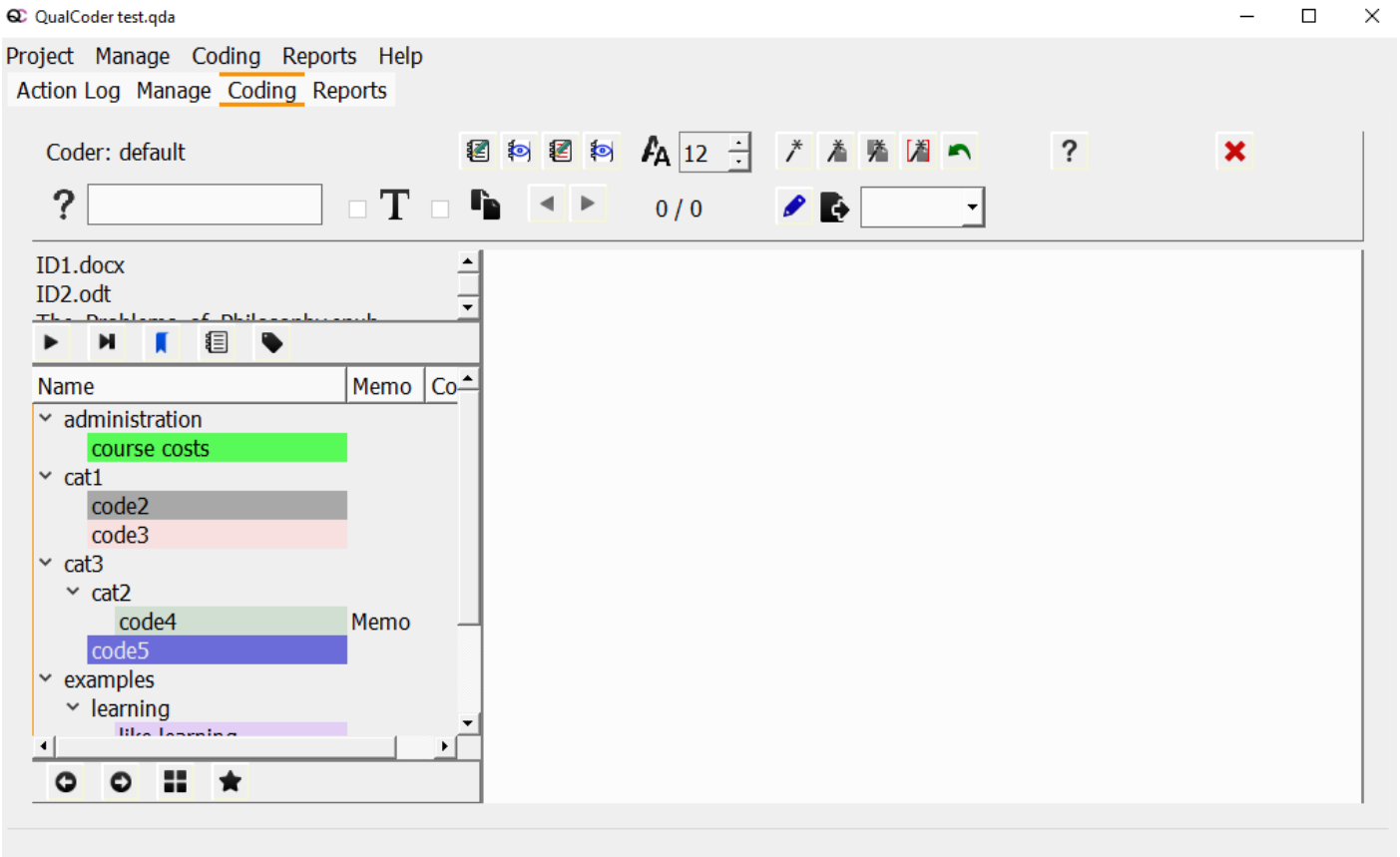


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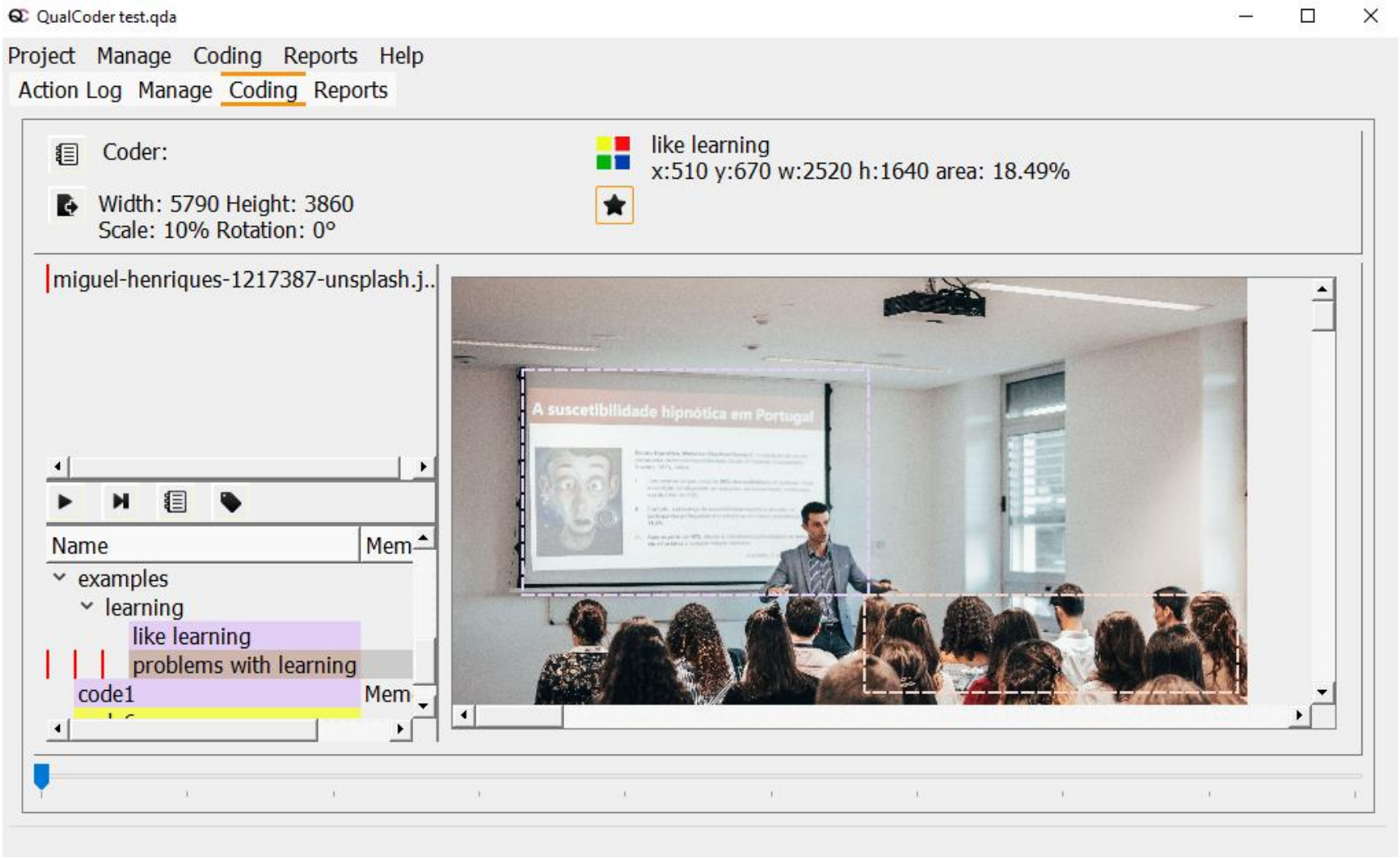


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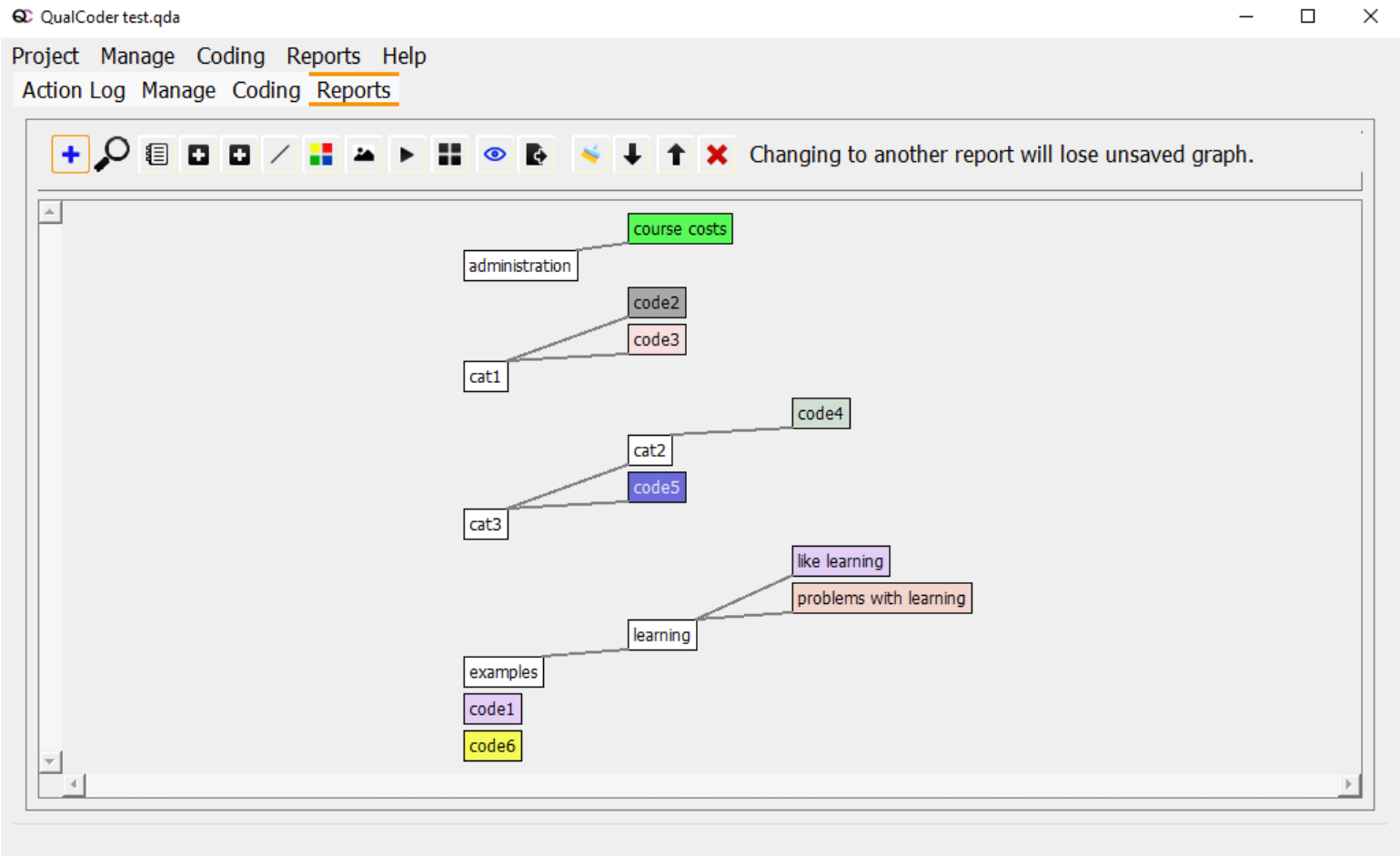
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Use of Qualitative Data Analysis Software Programs



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Q&A